A Computationally Tractable Model for Spatially Resolved Nuclide Concentrations in Flowing Fuel Molten Salt Reactors

Luke Seifert [1,*], Shayan Shahbazi[2], Scott Richards[2], Madicken Munk[1,3], Kathryn Huff[1]

[1]Advanced Reactors and Fuel Cycles Group, University of Illinois at Urbana-Champaign, Urbana, IL;
[2]Nuclear Science and Engineering Division, Argonne National Laboratory, Argonne, IL;
[3]School of Nuclear Science and Engineering, Oregon State University, Corvallis, OR

[leave space for DOI, which will be inserted by ANS]

ABSTRACT

Depletion simulations of flowing fuel molten salt reactors commonly do not distinguish between in-core and ex-core regions. Instead, they either irradiate the entire primary loop while scaling the flux, or irradiate the in-core volume alone, which leaves some amount of fuel salt out of the simulation. Either approximation results in some level of inaccuracy in nuclide concentrations. Various methods account for the flow of fuel salt between spatial regions, but a major hurdle is the computational cost. This work investigates a spatially resolved method to account for the impact flow and chemical removal has on concentrations of nuclides rapidly and accurately. Comparisons are made in this work between a spatially resolved method, spatially unresolved scaled flux method, other work from the literature, and experimental data. These comparisons show that the scaled flux method is sufficient for modeling most nuclides unless there is a large spatially dependent loss term, such as a large ex-core removal rate. In such a case, a spatially resolved method is needed.

Keywords: Molten Salt Reactor, Depletion, Chemical Removal, Molten Salt Reactor Experiment, Concentrations

1. INTRODUCTION

Liquid-fueled molten salt reactor (MSR) development started in the 1950s with the Aircraft Reactor Experiment [1]. Progress continued in the Molten Salt Reactor Experiment (MSRE) in the 1960s [2]. After that, solid-fueled light water reactors became the focus, which have since become the most common nuclear reactor used in the industry. However, the Generation IV International Forum in the early 2000s designated the MSR as one of the six advanced reactors of interest moving forward [3]. Since then, MSRs have garnered further interest, such as the 2020 Advanced Reactor Demonstration Program award for $113 million over seven years to the Molten Chloride Reactor Experiment.

Interest has shifted to MSRs, and many existing simulation codes and methods must adjust to account for the moving fuel. In particular, depletion simulations are of interest. In solid-fueled reactors, the fuel does not experience significant advection or diffusion of nuclides. However, MSRs have the fuel circulate through the in-core and ex-core regions. The depletion simulations must account for these regions with different power densities and the effects of circulation. Modeling this behavior accurately is vital, as the fuel composition affects reactor dynamics, reactor safety, and safeguards.

*seifert5@illinois.edu

One approach is computationally inexpensive and is called the scaled flux method [4]. The scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. This approach does not model the in-core and ex-core regions explicitly.

Alternatively, another approach has shown how spatial resolution can be investigated using a system of ordinary differential equations to solve the Bateman equation. This approach can estimate the steady-state concentration of certain nuclides across a flow loop, but it does not account for temporal resolution [5].

The toolkit MCNP/ADDER [6] has also been used to account for flow and chemical removal between neutron transport-coupled fuel depletion steps over the MSRE power history [7]. But within ADDER, ex-core loss terms (e.g., decay and chemical removal) are only treated after the calculation of the coarse depletion steps (e.g., months), as opposed to simultaneously on the timescale of the flow loop residence time (e.g., seconds). This results in an accurate removal of elements except for those nuclides that have both an ex-core loss term and a non-negligible neutron absorption cross section.

Alternatively, a more accurate treatment was recently demonstrated through MOOSE-based coupling of the deterministic neutron transport code Griffin with the thermal fluids code Pronghorn, allowing for spatial and temporal resolution of certain nuclides [8]. Tano et al. present an approach where only insoluble, medium-lived, and high concentration nuclides are tracked with pre-computed spatial distributions, which only need to be updated if the medium-lived isotopes are not in equilibrium after the thermal-hydraulics perturbation.

These approaches are more computationally expensive than those that only require solving a system of ordinary differential equations. However, it has been shown that including circulation of the fuel salt in depletion directly affects the equilibrium concentration of ^{135}Xe by approximately 45% [8].

Tano et al. and Betzler have both described fully modeling advection in depletion as computationally intractable [8, 4]. This is because tracking all nuclides over space and time requires far more computational work than handling a traditional time dependent system of ordinary differential equations (ODEs). This spatially resolved system of partial differential equations (PDEs) requires accuracy in the short time scale flow effects. These small time steps cause multi-year depletion simulations to be exceedingly expensive.

However, Tano et al. showed that not all nuclides must be modeled to obtain accurate results [8]. The present work extends a previous steady-state approach and builds on this idea from Tano et al. This work isolates isobars of interest, and adjusts the steady-state approach by accounting for time resolution in the system of differential equations. High-fidelity thermal hydraulic simulations are computationally expensive, so instead, a simple advection model is used to capture the in-core and ex-core effects. With the simplifications implemented, this method can run on an laptop in minutes, rather than requiring a supercomputer allocation.

2. METHODS

The ODE form of the Bateman equation written in simplified terms for nuclide i, which only contains time dependence, is given as

$$\frac{dN_i(t)}{dt} = S_i(t) - \mu_i(t)N_i(t). \tag{1}$$

Equation (1) is used for the scaled flux method in this work. The source term, $S_i(t)$, is defined as

$$S_i(t) = \phi(t)Y_iN_f\sigma_{f,f} + \sum_{j\neq i}\lambda_jN_j(t)\gamma_{j\rightarrow i}. \tag{2}$$

In Equation (2), $\phi(t)$ is the time dependent neutron flux in $\frac{n}{cm^2 s}$, Y_i is the fission yield of nuclide i, N_f is the concentration of the fissile nuclide in $\frac{atoms}{cm^3}$, σ_f is the fission cross section of the fissile nuclide in cm^2, $lambda_j$ is the decay constant in s^{-1}, $N_j(t)$ is the time dependent cross section of nuclide j, and $\gamma_{j\rightarrow i}$ is the probability of decay from nuclide j into nuclide i. The loss term, $\mu_i(t)$, in Equation (1) is defined as

$$\mu_i(t) = \lambda_{decay,i} + \lambda_{chem,i} + \phi(t)\sigma_{c,i}. \tag{3}$$

In Equation (3), λ_{decay} is the decay constant for nuclide i in s^{-1}, $\lambda_{chem,i}$ is the chemical removal rate in s^{-1}, and $\sigma_{c,i}$ is the total capture cross section of nuclide i in cm^2. When one-dimensional spatial dependence is added to Equation (1), the PDE is created of the form

$$\frac{\partial N_i(z,t)}{\partial t} + v(z,t)\frac{\partial N_i(z,t)}{\partial z} = S_i(z,t) - \mu_i(z,t)N_i(z,t). \tag{4}$$

Equation (4) is used in this work to provide the spatially resolved concentrations for target nuclides. In Equations (4) and (1), N_i is the atom density of nuclide i in $\frac{atoms}{cm^3}$, t is time in seconds (s), v is the linear flow rate in $\frac{cm}{s}$, z is the linear distance traveled in cm, S_i is the source of nuclide i in $\frac{atoms}{cm^3 s}$, and μ_i is the loss of nuclide i in s^{-1}. The spatially resolved equation does not include diffusion of nuclides because molten salt reactors are an advection-dominated system [9]. The source term is comprised of decay sources and fission yields in this work. Sources of nuclide i from parasitic absorption are currently not considered, as the impact is assumed to be sufficiently smaller than the fission and decay parent sources. However, future work may want to include this source. The loss term includes losses from both decay and the neutron total capture cross section in the core region. The loss term can also include a chemical removal term resolved in both space and time for nuclide i.

To solve the coupled system of equations given in Equation (4) in a computationally efficient manner, several simplifications are made. The results do not account for changes in the fuel or evolution of the neutron energy spectra [10]. Instead, these terms are assumed to be constant, which can be a reasonable approximation if fresh fuel is added and fission product poisons are removed during online reprocessing. Another simplification in this work is the use of only one spatial dimension and uniform flow. This includes simplifications to the flow of the fuel salt, which can only change in time and the single spatial dimension. This assumption neglects turbulent effects or vortices, which have a larger presence in some reactor designs, such as the Molten Salt Fast Reactor (MSFR). For flow through channels, such as in the MSRE, this approximation is more reasonable.

2.1. Data

Custom scripts combined with OpenMC use the linear flow rate, average neutron energy, average neutron flux, in-core fraction of fuel salt, ex-core fraction of fuel salt, in-core fuel salt volume, ex-core fuel salt volume, and average temperature to generate the appropriate cross sections and terms in Equation (4). The solver uses the average temperature and energy to generate one-group cross sections from OpenMC.

The scripts also use OpenMC to access general nuclear data in ENDF-B/VII.1 [10, 11]. These data include fission yield fractions, decay chains, branching ratios, half-lives, and cross sections. These data are important for properly constructing the source and loss terms for each nuclide, while the decay chains are useful for constructing the relationships between them.

The results in this work use the data shown in Table I. The fissile concentration, neutron flux, nominal power, power history, and in-core fraction come from the MCNP/ADDER simulation of the MSRE [6, 7].

The other terms come from calculations using the data from the OpenMSR project [12].

Table I. Simulation parameters used in this work.

Parameter	Value	Units
Fissile Nuclide	^{235}U	-
Fissile Concentration	8.41×10^{19}	$\frac{atoms}{cm^3}$
Neutron Flux	8.4×10^{12}	$\frac{n}{cm^2 s}$
Neutron Energy	0.02533	eV
Salt Density	2.32556	$\frac{g}{cm^3}$
Temperature	294	K
Nominal Power	7.34	MW
Primary Loop Flow Length	608.06	cm
In-core Salt Fraction	27.2	%
Ex-core Salt Fraction	72.8	%
Volume	2.116111	m^3
Linear Flow Rate	21.75	$\frac{cm}{s}$

2.2. Solver

This work uses finite differencing upwind, or backwards, in space and explicit in time to generate a coupled system of PDEs. The upwind in space approach is used to maintain numerical stability [13]. Numerical stability requires the solver to maintain the Courant-Friedrichs-Lewy (CFL) condition given as

$$\frac{\Delta t v}{\Delta z} \leq 1. \tag{5}$$

Equation (5) shows that, as the spatial mesh, Δz, becomes finer or the flow rate, v, becomes larger, the time step, Δt, must become smaller, which means higher computational cost. This is because a smaller time step becomes necessary to capture the effects spatially when the fluid moves faster through a spatial region Δz. The algorithm solves the coupled system of PDEs is the manner shown in Algorithm 1.

Algorithm 1 Solver Pseudocode

```
while t < t_final do
    while i ≤ i_net do
        S_i ← S_i,new
        μ_i ← μ_i,new
        N_i ← N_i,new
        i ← i + 1
    end while
    t ← t + Δt
end while
```

Algorithm 1 shows that, while any number of nuclides can be included in the decay chain, the computational cost will increase directly with the net number of nuclides, i_{net}. Additionally, the cost scales with the time step, though the time step is directly limited by the CFL condition in Equation (5). The algorithm updates the source terms, S_i representing the production of each nuclide i before each subsequent concentration update,

N_i, as the decay parents contribute as a source term for nuclide i. The algorithm updates the loss rates, μ_i, of each nuclide i as well, which is not necessary assuming constant flux and cross sections; however, for power transients, changes occur to the fission source term and the loss term.

2.3. Scaled Flux

The scaled flux method is a scaling of the ODE shown in Equation (1) to imitate the spatial resolution of the solve given in Equation (4) [14]. The method modifies the source and loss terms based on the in-core and ex-core fractions of the primary loop. Equation (6) scales Equation (2) using the in-core fraction, f_{in}, giving the source term using the scaled flux approach as

$$S_i(t) = \phi(t) Y_i N_f \sigma_{f,f} f_{in} + \sum_{j \neq i} \lambda_j N_j(t) \gamma_{j \to i}. \tag{6}$$

Equation (7) scales Equation (3) using the in-core fraction, f_{in} and the ex-core fraction, f_{ex}, as

$$\mu_i(t) = \lambda_{decay,i} + \lambda_{chem,i} f_{ex} + \phi(t) \sigma_{c,i} f_{in}. \tag{7}$$

3. RESULTS

The results for this work include spatial and chain sensitivity studies, a comparison between spatially resolved and spatially unresolved approaches, and validation and verification. For the sensitivity studies, the MSRE is simulated for one hour, while the spatial comparison uses 1.25 days for constant power and an additional 100,000 seconds after the step power change. The validation and verification simulates approximately 652 days of the MSRE operation.

Although the spatial sensitivity study is presented prior to the chain sensitivity, the results of the chain sensitivity inform the spatial sensitivity study. The spatial sensitivity study uses five nuclides leading to ^{135}Xe to ensure any spatial effects caused by decay parents is captured.

3.1. Spatial Sensitivity

The spatial sensitivity study informs the other analyses what level of spatial resolution is required to generate sufficiently accurate results. This analysis compares the concentration of ^{135}Xe after one hour in the MSRE using the spatially resolved PDE method for various numbers of spatial nodes. Table II provides details on the cost and accuracy of each simulation. The computational cost scales as $O(n^2)$ with the number of spatial nodes, n. This is because a finer spatial mesh also requires a finer time stepping scheme, as shown in Equation (5).

The spatial sensitivity study shows that 200 spatial nodes offers a high level of accuracy with a reasonable computational cost. The chain sensitivity and spatial comparison studies use 200 spatial nodes. The MSRE validation uses 100 spatial nodes instead to keep computational cost lower due to the larger simulation time while maintaining accuracy of approximately 1%.

3.2. Chain Sensitivity

The chain sensitivity study informs the other analyses what number of nuclides to include in the decay chain. This study compares the calculated ^{135}Xe concentration after one hour for various numbers of decay parents for ^{135}Xe included in the decay chain. Table III shows how varying the number of decay parents for ^{135}Xe

Table II. Spatial sensitivity detailing how cost and accuracy vary based on the number of spatial nodes for ^{135}Xe.

Nodes	Cost [s]	Error [%]
5	0.12	46.292
10	0.29	10.087
100	6.39	- 0.737
200	20.4	1.091
500	111	0.001
1000	409	-

in the isobar affects the results and simulation time.

Table III. Comparison between cost and accuracy for varying numbers of nuclides in the 135 isobar.

Nuclides	Cost [s]	Error [%]
Xe	4.40	84.1
Xe, ^{m}Xe	8.30	58.4
Xe, ^{m}Xe, I	13.1	31.8
Xe, ^{m}Xe, I, Te	16.9	1.08
Xe, ^{m}Xe, I, Te, Sb	20.7	0.00
Xe, ^{m}Xe, I, Te, Sb, Sn	24.5	-

The cost scales as $O(n)$ for n nuclides. Five and six nuclides have the same inaccuracy, because ^{135}Sn does not decay into ^{135}Sb in the OpenMC chain file used, so not including ^{135}Sn yields the same result. The expected contribution of ^{135}Sn is small regardless, so the error would also be small if it were included. Using five nuclides in the 135 isobar leading to ^{135}Xe appears to yield accurate results while additional nuclides do not have a significant impact.

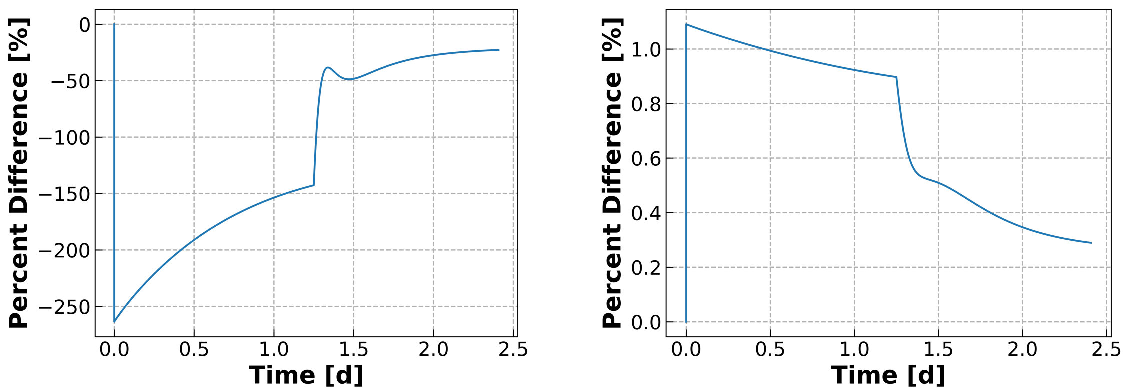
3.3. Spatial Comparison

The spatial comparison study compares the spatially resolved PDE to the scaled flux ODE method. In particular, a comparison with the scaled flux method yields insights into the difference between a spatially resolved and spatially unresolved method [14].

After 1.25 days, there is a difference of 0.91% when compared with the scaled flux method and 58.8% when compared with the ordinary differential equation method without scaling the flux. The difference of 58.8% is consistent with the difference of 44.24% found by Tano et al. which compares the unscaled flux to a quasi-static approach over 1.25 days for ^{135}Xe as well [8]. The difference found between this work and the Tano et al. work is likely due to the difference in the reactors, as Tano et al. investigate the MSFR while this work investigates the MSRE.

A step change in power determines if the spatially resolved method may be necessary to adequately handle sudden changes in reactor conditions. The power increases by an order of magnitude after 1.25 days. The percent difference for the scaled flux method decreases shortly after this power change to 0.3%. Figure 1 shows this, as it compares the unscaled and scaled flux percent differences over time for the average concentration of ^{135}Xe.

From these results, it appears that the scaled flux method is sufficient for modeling the concentrations of nuclides in MSRs even when there is a sudden change in power. However, the difference between the



(a) Unscaled flux **(b) Scaled flux**

Figure 1. Percent difference between the spatially resolved (PDE) method and the spatially unresolved (ODE) method on the ^{135}Xe concentration.

spatially resolved PDE and the non-spatially resolved ODE becomes larger when there is chemical removal in the ex-core region. There is an initial large positive difference in the concentrations which exponentially decreases to a positive steady-state difference.

Figure 2 shows how varying the cycle time affects the inaccuracy of the scaled flux method. The cycle time, T_{cyc}, describes the removal rate of xenon, for which a larger cycle time corresponds to a slower removal rate. Specifically, the removal rate, λ_r, is the inverse of the cycle time.

Figure 2 shows that, for cases where in which the removal rate is greater, the scaled flux method is less accurate than a spatially refined model. Figure 2 represents this both by the peak difference as well as the final difference, which occurs as the difference approaches an equilibrium value.

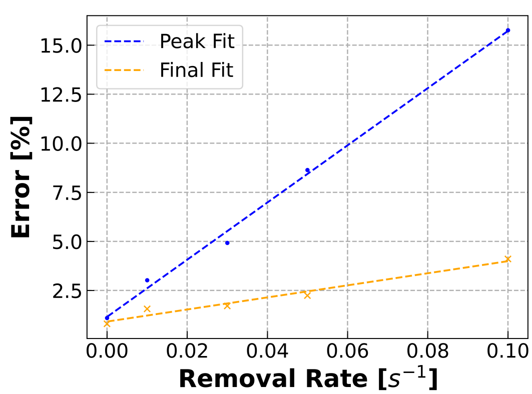


Figure 2. The errors from using the scaled flux method to handle spatially resolved chemical removal for various chemical removal rates of xenon.

To simulate chemical removal for the other elements in the decay chain, the cycle time of 20 seconds is used and then applied to the elements in the 135 isobar chain leading to xenon. Table IV shows the results of this study.

Table IV. The inaccuracies of using the scaled flux method to handle spatially resolved chemical removal for nuclides in the 135 isobar chain leading to xenon.

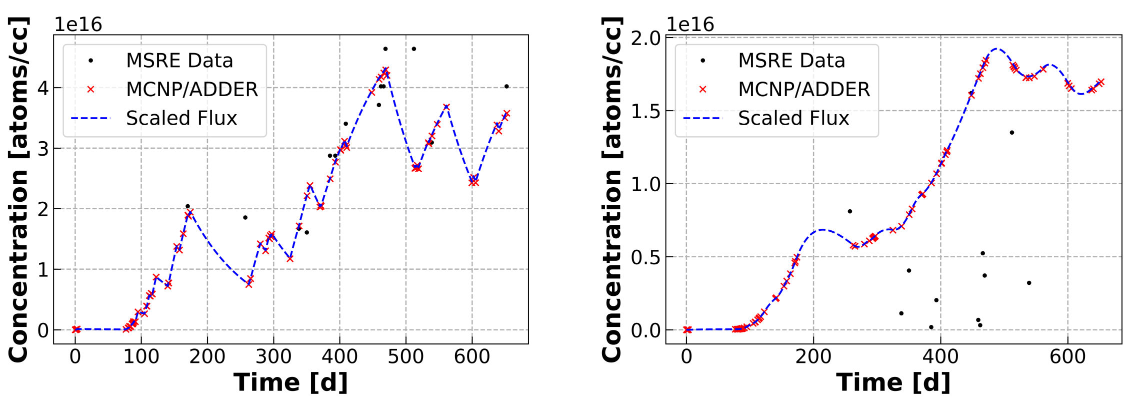
Removed Elements	Peak Diff [%]	Final Diff [%]
-	1.1	0.8
Xe	8.6	2.2
Xe, I	8.7	3.9
Xe, I, Te	8.7	4.0
Xe, I, Te, Sb	8.7	4.0

3.4. Validation and Verification

This work uses the experimentally measured MSRE ladle salt samples over time during uranium-235 operation for validation [15]. The validation uses concentrations for ^{95}Zr and ^{95}Nb during operation in the MSRE from January 24th, 1966 to November 7th, 1967. ^{95}Zr is a salt-seeking nuclide without an appreciable noble gas precursor, which means it can be modeled without having to account for chemical removal. As discussed previously, the scaled flux method is sufficient for modeling the spatially resolved depletion without chemical removal. Not only is there a spatial component that does not need to be modeled, but the spatial resolution is also not required. This means only the decay chain sensitivity needs to be included, which has linear scaling.

For analyzing the noble metal ^{95}Nb, this work considers chemical removal in the ex-core region. The chemical removal rate used for ^{95}Nb in this work is the value from Shahbazi et al. of $2.3 \times 10^{-8} s^{-1}$ [16]. This value accounts for the removal of elements from the fuel salt to the cover gas in the pump bowl. This removal rate is small because it is meant to only represent the removal of niobium from the salt to the off-gas system and does not include removal via deposition throughout the primary loop. Therefore, because it is small relative to the decay constant, the scaled flux method is sufficient for modeling both the ^{95}Zr and ^{95}Nb concentrations.

Figure 3 shows the concentrations of ^{95}Zr and ^{95}Nb over time. In the figure, comparisons are made between the experimental results of the MSRE [15], analysis using MCNP/ADDER [7], and the current work using the scaled flux method.



(a) ^{95}Zr concentration **(b) ^{95}Nb concentration**

Figure 3. Concentrations over time in the Molten Salt Reactor Experiment.

There appears to be reasonable agreement between the current work and the results from MCNP/ADDER, which does not account for chemical removal. The agreement in ^{95}Nb is likely due to the very small removal rate the scaled flux method uses. However, the MSRE experimental data does not agree at each point. This can be explained by uncertainty in the measurements, chemical effects which aren't modeled in MCNP/ADDER or this work, and other minor contributing factors these models do not include. Particularly, the ^{95}Nb concentrations from the MSRE measurements are smaller than the model predicts, which means that a larger chemical removal rate would likely fit these data better.

4. CONCLUSIONS

This work demonstrates a simplified approach to calculating spatially resolved nuclide concentrations in molten salt reactors, offering verification and preliminary validation of the results. The results offer close agreement with MCNP/ADDER, while the data from the MSRE for salt-seeking isotopes is in reasonable agreement. The scaled flux method was demonstrated to be effective for modeling nuclide concentrations with small chemical removal rates, as the error scales linearly with increasing chemical removal rates. What this means practically is that most elements modeled in MSRs can be treated using the scaled flux method, while those with large chemical removal rates should be resolved spatially. Future work will involve continued verification and validation efforts for isotopes that exhibit in-core and ex-core loss terms due to neutron capture and chemical removal, respectively.

ACKNOWLEDGEMENTS

Luke Seifert: Methodology, formal analysis, writing - original draft. **Shayan Shahbazi**: Conceptualization, methodology, resources, supervision, writing - review and editing. **Scott Richards**: Conceptualization, supervision. **Madicken Munk**: Supervision, funding acquisition, writing - review and editing. **Kathryn Huff**: Supervision, funding acquisition, writing - review and editing. Additional thanks to Samuel Dotson for review of the scripts used in this work and Nathan Ryan for review of this paper.

REFERENCES

[1] E. S. Bettis, W. B. Cottrell, E. R. Mann, J. L. Meem, and G. D. Whitman. "The Aircraft Reactor Experiment—Operation." Nuclear Science and Engineering, **volume 2**(6), pp. 841–853 (1957). URL https://doi.org/10.13182/NSE57-A35497. Publisher: Taylor & Francis _eprint: https://doi.org/10.13182/NSE57-A35497.

[2] P. N. Haubenreich and J. R. Engel. "Experience with the Molten-Salt Reactor Experiment." Nuclear Applications and Technology, **volume 8**(2), pp. 118–136 (1970). URL https://doi.org/10.13182/NT8-2-118. Publisher: Taylor & Francis _eprint: https://doi.org/10.13182/NT8-2-118.

[3] J. E. Kelly. "Generation IV International Forum: A decade of progress through international cooperation." Progress in Nuclear Energy, **volume 77**, pp. 240–246 (2014). URL https://www.sciencedirect.com/science/article/pii/S0149197014000419.

[4] B. Betzler. "Liquid-fueled Molten Salt Reactor Depletion Modeling." Technical report, Oak Ridge National Lab. (ORNL), Oak Ridge, TN (United States) (2021). URL https://www.osti.gov/biblio/1809981. 2.

[5] S. Shahbazi, P. Romano, T. Fei, and D. Grabaskas. "Steady-state radiochemical transport model of the molten salt reactor experiment." Journal of Radioanalytical and Nuclear Chemistry, **volume 331**(12), pp. 5247–5257 (2022). URL https://doi.org/10.1007/s10967-022-08535-3. 4.

[6] K. Anderson, V. Mascolino, and A. G. Nelson. "User Guide to the Advanced Dimensional Depletion for Engineering of Reactors (ADDER) Software (V.1.01)." Technical Report ANL/RTR/TM-21/8-Rev.1, Argonne National Laboratory (ANL), Argonne, IL (United States) (2022). URL https://www.osti.gov/biblio/1866062.

[7] T. Fei, S. Shahbazi, J. Fang, and D. Shaver. "Validation of NEAMS Tools Using MSRE Data." Technical Report ANL/NSE-22/48, Argonne National Laboratory (ANL), Argonne, IL (United States) (2022). URL https://www.osti.gov/biblio/1880993.

[8] M. Tano, S. A. Walker, and A. Abou-Jaoude. "A Coupled Griffin and Pronghorn Method for Spatially Resolved Depletion Analysis of Molten Salt Reactors." In American Nuclear Society Mathematics & Computation (2023).

[9] S. M. Park and M. Munk. "Verification of moltres for multiphysics simulations of fast-spectrum molten salt reactors." Annals of Nuclear Energy, **volume 173**, p. 109111 (2022). URL https://www.sciencedirect.com/science/article/pii/S0306454922001463.

[10] P. K. Romano, N. E. Horelik, B. R. Herman, A. G. Nelson, B. Forget, and K. Smith. "OpenMC: A state-of-the-art Monte Carlo code for research and development." Annals of Nuclear Energy, **volume 82**, pp. 90–97 (2015). URL https://www.sciencedirect.com/science/article/pii/S030645491400379X. Joint International Conference on Supercomputing in Nuclear Applications and Monte Carlo 2013, SNA + MC 2013. Pluri- and Trans-disciplinarity, Towards New Modeling and Numerical Simulation Paradigms.

[11] M. Chadwick, M. Herman, P. Obložinský, M. Dunn, Y. Danon, A. Kahler, D. Smith, B. Pritychenko, G. Arbanas, R. Arcilla, R. Brewer, D. Brown, R. Capote, A. Carlson, Y. Cho, H. Derrien, K. Guber, G. Hale, S. Hoblit, S. Holloway, T. Johnson, T. Kawano, B. Kiedrowski, H. Kim, S. Kunieda, N. Larson, L. Leal, J. Lestone, R. Little, E. McCutchan, R. MacFarlane, M. MacInnes, C. Mattoon, R. McKnight, S. Mughabghab, G. Nobre, G. Palmiotti, A. Palumbo, M. Pigni, V. Pronyaev, R. Sayer, A. Sonzogni, N. Summers, P. Talou, I. Thompson, A. Trkov, R. Vogt, S. van der Marck, A. Wallner, M. White, D. Wiarda, and P. Young. "ENDF/B-VII.1 Nuclear Data for Science and Technology: Cross Sections, Covariances, Fission Product Yields and Decay Data." Nuclear Data Sheets, **volume 112**(12), pp. 2887 – 2996 (2011). URL https://www.sciencedirect.com/science/article/pii/S009037521100113X. Special Issue on ENDF/B-VII.1 Library.

[12] L. Chierici, A. Stubsgaard, L. Labrie-Cleary, M. Akner, and E. Knudsen. "Open Source Software for Modeling Molten Salt Reactors." https://github.com/openmsr/msre (2023).

[13] S. Linge and H. P. Langtangen. Advection-Dominated Equations, pp. 323–351. Springer International Publishing, Cham (2017). URL https://doi.org/10.1007/978-3-319-55456-3_4.

[14] B. Betzler. "Liquid-fueled molten salt reactor depletion modeling." Technical report, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States) (2021).

[15] E. Compere, S. Kirslis, E. Bohlmann, F. Blankenship, and W. Grimes. "Fission product behavior in the molten salt reactor experiment." Technical report, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States) (1975).

[16] S. Shahbazi, S. Walker, M. Tano, T. Fei, Y. Cao, and D. Grabaskas. "Modeling Radionuclide Inventories in MSR Off-Gas Systems with Radiochemical Transport Analysis." Technical report, Argonne National Laboratory (ANL), Argonne, IL (United States) (2024).